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REGULATION OF BLOOD PRESSURE

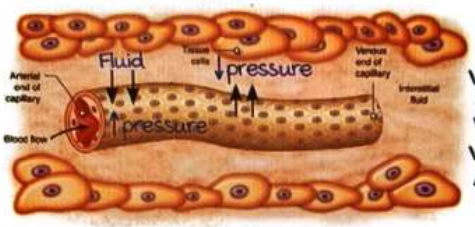
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Blood pressure : Single most important factor regulating the CVS.

MECHANISMS :

1. Short term :
   - CNS controlled.
   - Rapid acting.
     a. Baroreflex.
     b. Chemoreflex.
     c. CNS ischemic response.

2. Intermediate term :
   Activation within hours.



Vasodilation : ↓pressure
Vasoconstriction : ↑pressure

   a. Capillary fluid shift.
   b. Stress relaxation.
   c. Renin Angiotensin Aldosterone system (RAAS).

3. Long term :
   - Activation within days.
   - Regulation by kidney (Pressure diuresis)

   coregulates fluid volume & urine output.

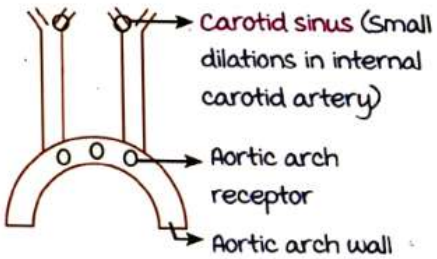
Short Term Regulation

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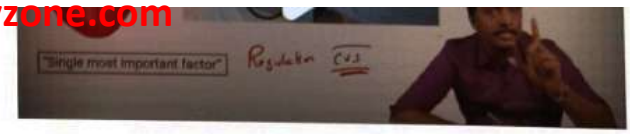
BAROREFLEX :

↑BP via baroreflex → (Negative feedback) ↓BP

Mechanism	
Stimulus	↑BP
Receptor	Baroreceptor (Distension)
Afferent (Buffer/sinus nerves)	• Carotid (CN IX : Hering nerve) • Aortic (CN X : Cyon's nerve)
Centre	Nucleus Tractus Solitarius (in medulla)
Efferent	Glutamate → Caudal ventrolateral medulla (CVLM) → GABA → Rostral ventrolateral medulla (RVLM) → Sympathetic inhibition; Glutamate → Nucleus ambigous → Parasympathetic activation
Effector	Heart & blood vessel
Response	• ↓BP. • ↓HR.



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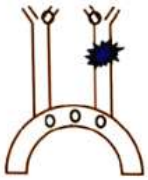
Note :

marey's law : ↑ BP leads to reflex ↓ in HR $\left(BP \times \frac{1}{HR}\right)$

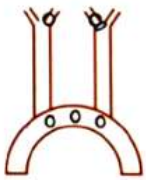
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Experiments :

- Occlusion of common carotid artery.
- Clamping below carotid sinus.



↓ Blood flow to internal carotid artery
↓ Pressure : Baroreflex inhibited
↑ BP & ↑ HR (↑ Sympathetic activation)



- Clamping above carotid sinus.
  ↑ pressure : Baroreflex activated.
  ↓ BP & ↓ HR (↑ Sympathetic inhibition).

Clinical aspects :

1.



Supine



On standing immediately

↓ BP (Baroreflex inhibited)

↓

SNS activated
Release of norepinephrine

↑ HR

Vasoconstriction (Via α-receptors)

↓

↑ Total peripheral resistance

if SNS underactivity

↓

Fall d/t postural hypotension

2. Carotid Sinus Massaging/Czermak-Hering test.



Carotid sinus massaging

Activates Baroreflex → ↓HR

- Used in Rx of Paroxysmal Supraventricular tachyarrhythmia (PSVT).

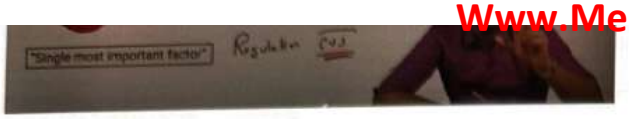
Other baroreceptors :

Atrial & Pulmonary artery baroreceptors.

- Low pressure receptor (Volume sensing receptor).
- 2 types of atrial stretch receptors :
  Type A : Activated during atrial systole.
  Type B : Activated during atrial diastole.

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REGULATION AT DIFFERENT LEVELS OF BP :

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Baroreceptors :

- 1st line to regulate BP.
- Stimulus : mean arterial pressure (MAP) of 70 – 140 mmHg.
  Response : up to MAP of 180 mmHg.

Chemoreceptors :

Second Line for BP control.

Stimulus : MAP of 40–70mmHg.

Chemoreflex → Direct effect → Activates PNS (↓HR)
Chemoreflex → Indirect effect → Activates SNS (↑HR)
} Effect on heart rate Variable

CNS Ischemic Response :

Last line of control/last ditch stand mechanism.

Stimulus : MAP ≤ 40mmHg.

Response : ↓ Cerebral perfusion (D/t low perfusion pressure).

↓

Activation of RVLM (SNS).

↓

↑↑ MAP (as high as 250 mmHg) (most powerful).

BP Waves

00:42:13

Mayer waves/Vasomotor waves :

- Oscillation of Nervous control mechanisms.

Traube-Hering waves :

- Fluctuations in BP synchronised with respiration.

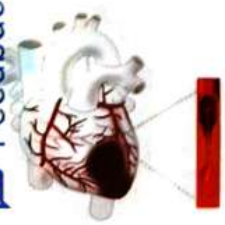
During inspiration :
↓ Systolic blood pressure

During early part of expiration :
↑ Arterial pressure

Cardiac Reflexes

00:44:25

1. Bezold-Jarisch reflex :

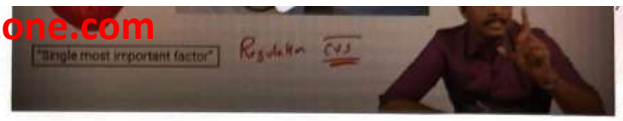


myocardial infarction

Serotonin from injured tissue → Activates Bezold Jarisch reflex → Bradycardia (↓HR) & Hypotension

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2. Cushing's reflex :

Head injury ⟶ ↑ ICT ⟶ Bradycardia (↓HR)

3. Bainbridge reflex :

Ex : Congestive Cardiac Failure ⟶ Fluid overload ⟶ Activates SA node (Right atrium enlargement) ⟶ Tachycardia (↑HR)

Chemical Regulation

00:48:50

1. Vasoconstrictors :

Mechanism : ↑ BP

Examples :

- Urotensin (most potent).
- Norepinephrine (α agonist).
- Endothelin (ET) : Produced from endothelial cell of lungs.

  ETA receptor ETB receptor

  ↓

  Endothelin I Overactivity

  ↓

  Pulmonary artery hypertension ⟶ Rx : Bosentan (Endothelin blocker).
- Vasopressin (V1 receptor) : ↓ Blood Flow.

  Eg : Terlipressin : Used in Rx of esophageal varices.
- Angiotensin II.

2. Vasodilators :

Mechanism : ↓ BP.

- Used in treatment of hypertension.

Examples :

- Calcitonin G related peptide (CGRP) : most potent.
- Nitric oxide (2nd messenger: cGMP) : Used in Rx of angina.
- Natriuretic peptide : ANP, BNP, CNP.
  - 2nd messenger : cGMP.
  - Nesiritide : BNP analogue.
- Bradykinin.

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GASTROINTESTINAL SECRETIONS

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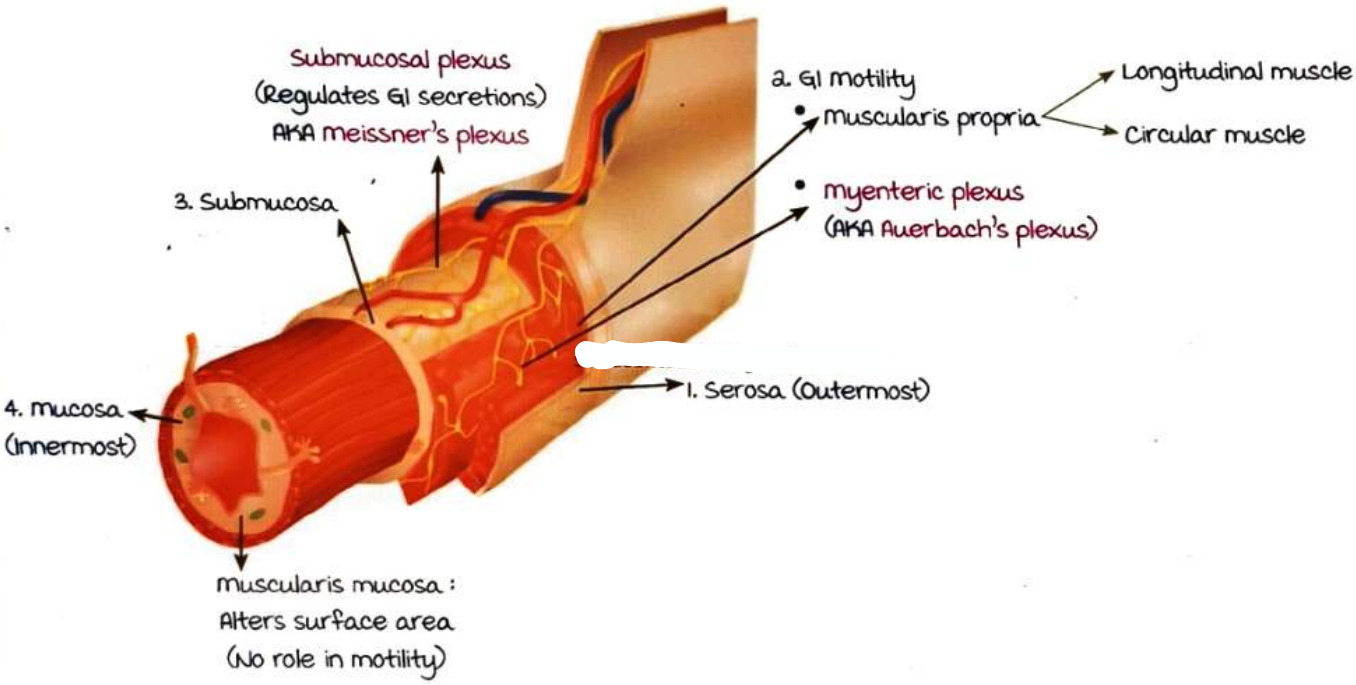
Introduction

00:00:26

Digestion & absorption of food requires :

- GI secretions (With enzymes) ← regulated by → Enteric nervous system.
- GI motility ← regulated by → GI hormones.

Layers of GI tract

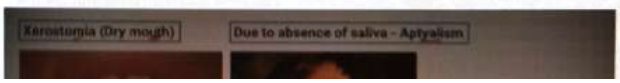


Daily water turnover in GIT :

Total water in GIT : 9000 mL/day $\xrightarrow{\text{Reabsorption}}$ 8800 mL/day $\xrightarrow[\text{In stool}]{\text{Excretion}}$ 200 mL.

- Ingested fluids : 2000 mL/day.
- GI secretions :
  - Saliva : 1500 mL/day.
  - Stomach : 2500 mL/day.
  - Exocrine pancreas : 1500 mL/day.
  - Bile : 500 mL/day.
  - Intestine : 1000 mL/day.

- Jejunum : 5500 mL.
- Ileum : 2000 mL.
- Colon : 1300 mL.



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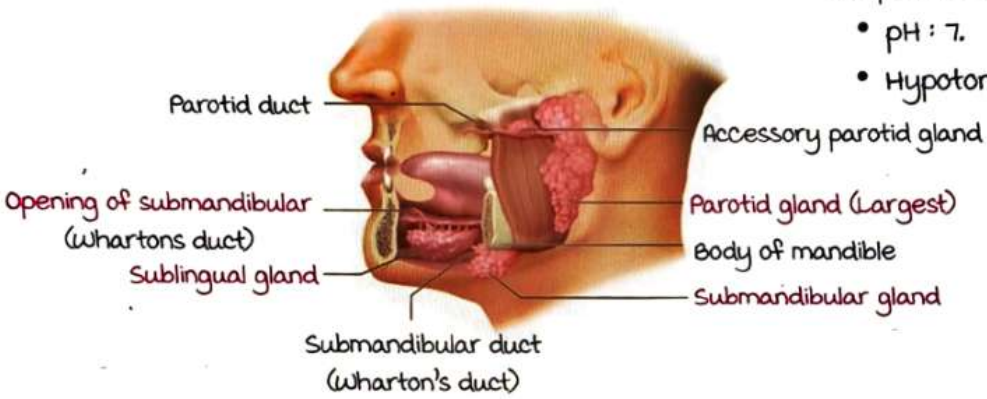
Saliva

00:11:09

Digestive & protective functions.

Salivary glands :

Properties of saliva :
- pH : 7.
- Hypotonic to plasma.



3 major salivary glands

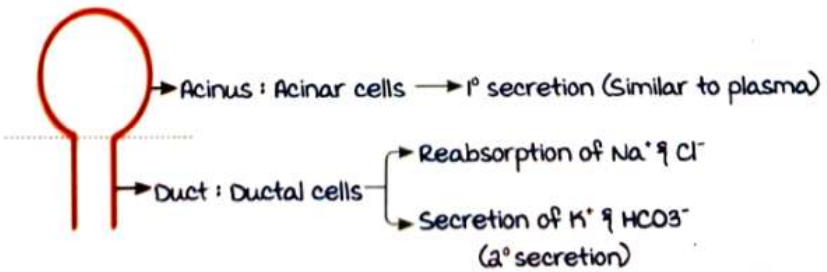
2 minor salivary glands :
- Weber's gland (Tonsils).
- Von Ebner's gland (Base of tongue).

FUNCTIONS OF SALIVA :

Function	Details		
Digestion of	Carbohydrate	Fats	Protein
Enzyme involved in digestion	Salivary amylase/ptyalin	Lingual lipase (By von ebners glands)	No enzymes
Lubrication	To prevent injury to buccal mucosa by dry foods.		
Protection	Immunity conferred through • Secretory IgA. • Lactoferrin } Anti-bacterial • Lysozymes } Anti-bacterial		

Salivon :

Functional unit of salivary gland.



- During high flow rates, ↓ time for resorption → Secretion has ↑ Na^+ and Cl^-.

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Regulation of secretion :

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Salivation is an unconditioned reflex.

a. Regulation by autonomic nervous system

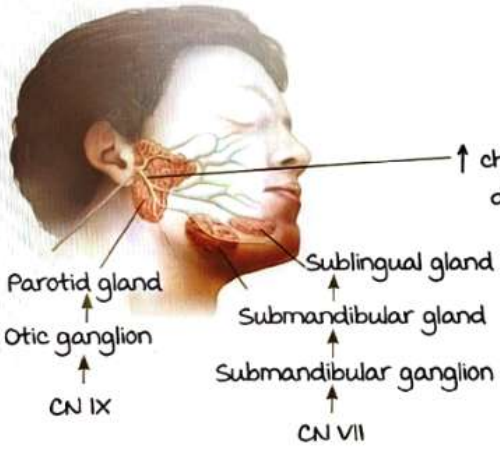
- **Sympathetic system**
  - ↑ in secretion of thick saliva
- **Parasympathetic system**
  - Profound ↑ in secretion of thin watery saliva.

b. Bradykinin → Vasodilation → ↑ blood flow to salivary gland → ↑ secretion.

Physiological factors that affect saliva secretion :

- Increase : Sight, smell or thought of food.
- Decrease : Sleep, fear & fatigue.

Innervation of salivary glands :



Abnormalities of salivary secretions :

- **Saliva**
  - **Absent secretion** → Aptyalism
  - **↓ secretion** → Xerostomia (Dry mouth)

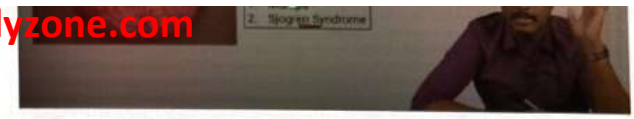


Commonly seen in :
- mumps.
- Sjogren syndrome.

  - **Associated abnormalities** → Injury to facial nerve during parotid surgery → During regeneration
    - Innervates sweat glands → ↑ salivation → ↑ sweating
    - Innervates lacrimal gland → ↑ salivation → ↑ lacrimation (Crocodile tears/gustatory lacrimation)

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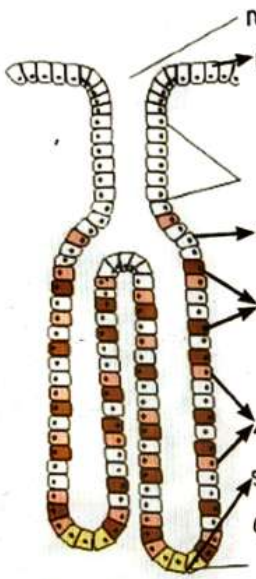
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Gastric Secretions

00:28:41

- Volume : 2.5 L/day.
- pH : 1.5-3 (most acidic secretions).

Gastric glands



Mucosal surface

1. Surface cell : Produces
   - HCO_3^- rich mucus for acid neutralisation.
   - Trifoil peptides : Confers immunity.
2. Neck cells : Stem cells of gastric glands.
3. Parietal cells : Produces
   - HCl acid.
   - Intrinsic Factor (IF) of Castle : For Vit B_{12} absorption.
4. Enterochromaffin-like cells (ECL) : Produces histamine → ↑ acid secretion
5. Chief cells : Produce pepsinogen enzyme.
6. Antral gastric glands : G cells → Gastrin → ↑ acid secretion.

Flask shaped gastric glands

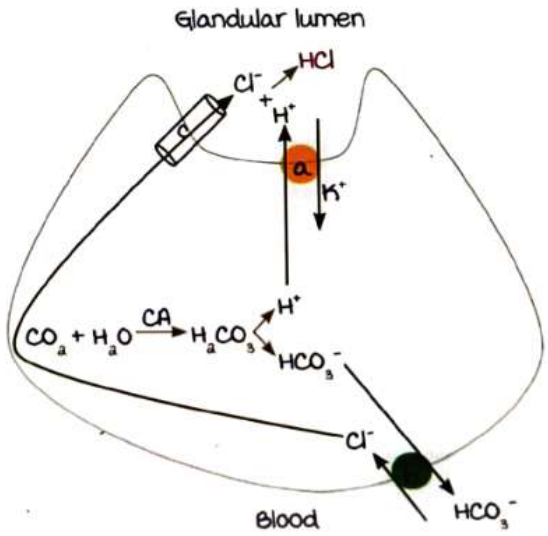
Note :

- In Pernicious anemia :

Autoantibodies to parietal cells → Destruction of parietal cells → Achlorhydria (↓ acid secretion) + Anemia (↓ Vit B12 absorption)

Destruction of parietal cells → Atrophic gastric mucosa

Acid production in parietal cell :



- a : H^+/K^+ ATPase proton pump.
- b : Cl^-/HCO_3^- exchanger.
- c : Cl^- channel.
- CA : Carbonic anhydrase.

Post-prandial alkaline tide :

↑ in HCl production after meal

↓

↑ HCO_3^- to blood (Post-prandial alkaline tide).

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Regulation of gastric acid secretion :

----- Active space -----

Factors that increase :

Substance	Source	Acts on (Receptor)
Histamine	ECL cells	H_2
Gastrin	Antral G-cells	Cholecystokinin-B (CCK-B)
Acetyl choline	Vagal synapses	M_3
Coffee	Ingestion	-
Alcohol		-
Ca^{2+}		-

Factors that decrease :

- Excess acid : Negative feedback.
- Somatostatin : Universal inhibitor.
- Prostaglandin.
- Cholecystokinin.
- Secretin.
- Calcitonin.

Functions of gastric secretion :

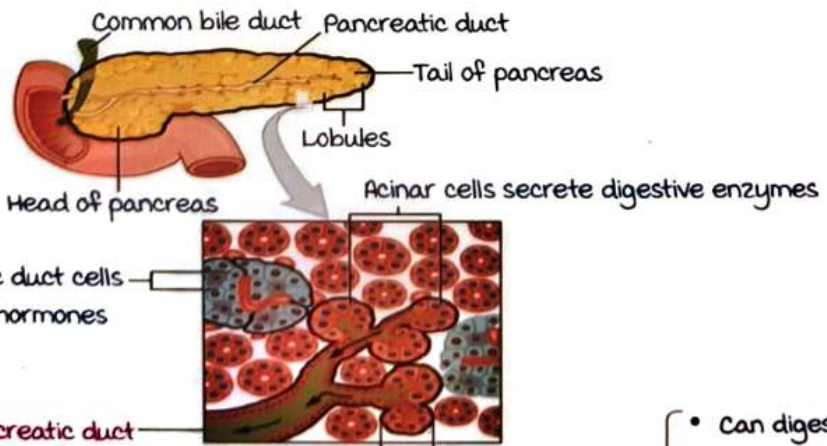
Function	Details		
Digestion of	Carbohydrate	Fats	Proteins
Enzyme involved	No enzymes	Gastric lipases	• Pepsinogen $\xrightarrow{\text{Activated by HCl}}$ Pepsin • Rennin (In cows)
Protection	↑ acidity → microorganism killed		

Phases of gastric acid secretion :

	Cephalic phase	Gastric phase	Intestinal phase
% of gastic acid secretion	20-30%	60-70%	10%
Gastric secretion occurs	In absence of food	When food enters stomach	D/t products of protein digestion
Initiated by	Sight, smell or thought of food	Antral G-cells → Gastrin → ↑ acid secretion	Intestinal G-cells → Gastrin → ↑ acid secretion

Exocrine Pancreatic Secretion

00:49:53



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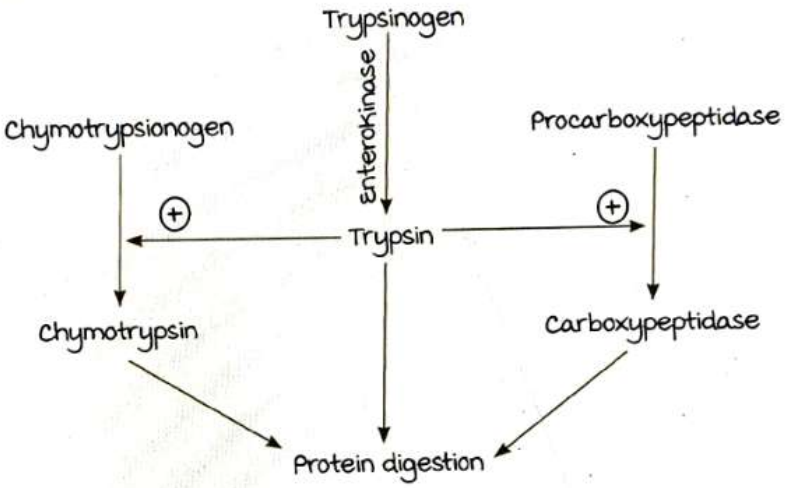
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Function :

Digestion of

- **Carbohydrates** — Pancreatic amylase
- **Fats** — Pancreatic lipase, assisted by co-lipase
- **Proteins**
  - Zymogens (Inactive enzymes).
  - Trypsinogen.
  - Chymotrypsinogen.
  - Procarboxypeptidase.

Activation of zymogens in small intestine :



Note :

Inflammatory process → Premature activation of zymogens → Acute pancreatitis (Auto-digestion).

Regulation :

Hormonal regulation

- **Cholecystokinin-pancreozymin (CCK-PZ)**
  - Constriction of gall bladder.
  - ↑ pancreatic enzymes in secretion.
- **Secretin (Nature's antacid)**
  - Released from S-cells in intestine on stimulation by acid chyme
  - ↑ HCO_3^- secretion into pancreatic juice
  - Alkalinity of pancreatic secretion for acid neutralisation

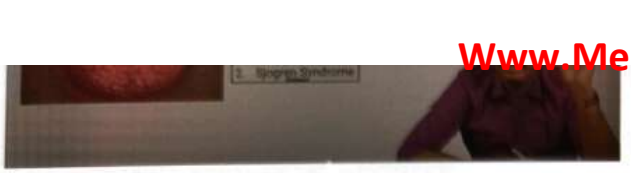
Note :

Secretin :

- First hormone to be discovered.
- ↓↓ enzyme concentration.

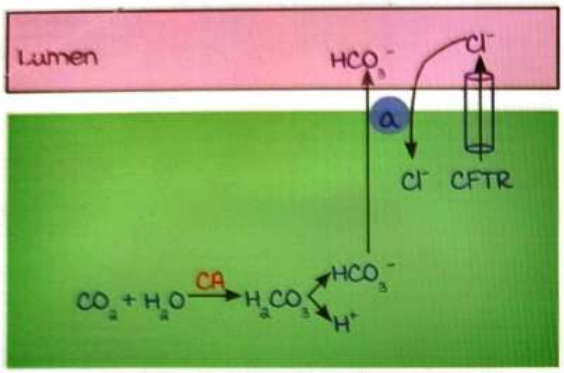
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mechanism of HCO_3^- secretion :

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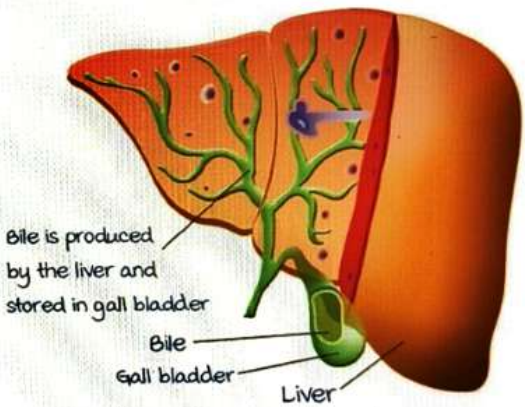
Pancreatic cell

- a : Cl^-/HCO_3^- exchanger.
- CFTR : Cystic fibrosis transmembrane conductance regulator → Chloride channel opening (mutation : Cystic fibrosis).

Biliary Secretion

01:03:51

Liver : Synthesis of bile (97% water) → Gall bladder : Storage of concentrated bile (87% water).



Biliary secretion

Functions of bile :

- Digestion and absorption of lipids.
- Excretion of :
  - Bile pigments.
  - Cholesterol.

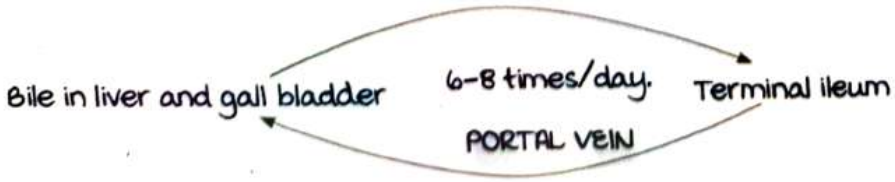
Constituents of bile

Primary bile acids (In liver)
- Cholic acid.
- Chenodeoxycholic acid.

→ Intestinal flora → Secondary bile acids (In intestine)
(Eg : Deoxycholic acid, Lithocholic acid)
↓
Conjugation in liver (With glycine/taurines)
↓ + Na^+
Bile salts (Sodium salts)
(Eg : Sodium glycocholate salts)

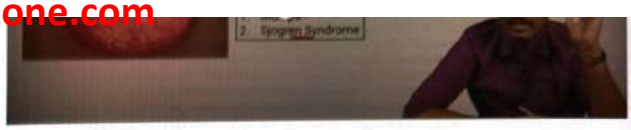
Enterohepatic circulation :

Purpose : Minimize energy consumption during bile synthesis.



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Factors affecting bile synthesis :

- Choleretic (↑ bile synthesis)
  - Bile salt.
  - Secretin.
  - Vagal stimulation.
- Cholagogues (↑ contraction of gall bladder)
  - CCK.

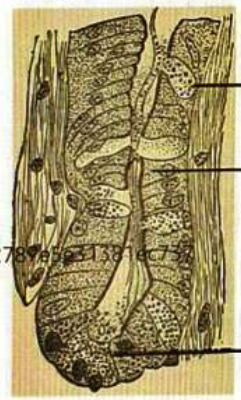
Intestinal Secretions

01:12:22

AKA succus entericus.

Site : Crypts of Lieberkuhn.

Types of cells :



1. Goblet cell : Produces mucus (For acid neutralisation)
2. Epithelial cell : Source of succus entericus
3. Paneth cell (Base) : ↑ concentration of zinc
   - Produce → Defensins, Lysozymes } Anti-bacterial

Note :

Maximum K^+ secretion : Saliva

Maximum K^+ concentration : Colonic secretion } measured in mEq/L.

Feedback